MACHINE LEARNING

LP-III Practical Viva | 100 Questions with Detailed Answers

How to use this question bank: Questions marked with ★ are high-priority questions commonly asked in practical vivas. The remaining questions give a mixture of basic definitions, conceptual questions, model evaluation, practical implementation and deeper questions. Answers are written in simple language and are generally 3–4 lines long.

Practical coverage: Uber fare prediction, email spam classification, Gradient Descent, KNN on the diabetes dataset and K-Means clustering. Theory coverage includes ML fundamentals, preprocessing, regression, classification, clustering and model evaluation.

1. Machine Learning Fundamentals

1. What is Machine Learning? ★

Answer: Machine Learning is a branch of Artificial Intelligence in which computers learn patterns from data and use them to make predictions or decisions. Instead of writing every rule manually, we train a model using examples. The model can then make predictions on new data.

2. How is Machine Learning different from traditional programming?

Answer: In traditional programming, a programmer writes rules and the computer uses them with input data to produce output. In Machine Learning, the computer learns patterns from training data and expected outputs. The learned model is then used to predict outputs for new inputs.

3. Differentiate AI, ML and Deep Learning. ★

Answer: Artificial Intelligence is the broad field of making machines perform tasks that normally require intelligence. Machine Learning is a subset of AI that learns patterns from data. Deep Learning is a subset of ML that uses neural networks with multiple layers.

4. What are features and labels? ★

Answer: Features are the input columns used by a model to make a prediction. The label, also called the target, is the output that the model is expected to predict. For example, in fare prediction, trip distance may be a feature and fare amount may be the target.

5. What is a dataset?

Answer: A dataset is a collection of examples or observations used for analysis and model training. Usually, rows represent individual observations and columns represent features. In supervised learning, one column may contain the target label.

6. What is supervised learning? ★

Answer: Supervised learning trains a model using labelled data, where the correct output is already known. The model learns the relationship between input features and output labels. Regression and classification are common supervised learning tasks.

7. What is unsupervised learning? ★

Answer: Unsupervised learning works with data that does not have target labels. The algorithm tries to discover patterns, groups or structures in the data. K-Means clustering is a common example.

8. What is semi-supervised learning?

Answer: Semi-supervised learning uses a small amount of labelled data together with a larger amount of unlabelled data. It is useful when collecting labels is expensive or time-consuming. The unlabelled examples help the model learn the structure of the data.

9. What is reinforcement learning?

Answer: In reinforcement learning, an agent interacts with an environment and takes actions. It receives rewards or penalties based on those actions and learns a strategy to maximize long-term reward. It is used in robotics, games and decision-making systems.

10. Differentiate classification and regression. ★

Answer: Classification predicts a category, such as spam or not spam, or diabetic or non-diabetic. Regression predicts a continuous numerical value, such as an Uber fare or house price. The target variable determines which type of problem it is.

2. Data Preprocessing and Feature Engineering

11. Why is data preprocessing required? ★

Answer: Real-world data may contain missing values, duplicate records, incorrect entries, noise and features with different scales. Preprocessing improves data quality before training. Poor-quality input data can lead to inaccurate or unreliable models.

12. What is a missing value and how can it be handled?

Answer: A missing value is an entry that is not available in the dataset. It can be handled by removing affected rows or columns, or by filling values using mean, median or mode. The method depends on the feature and the amount of missing data.

13. What is feature scaling? ★

Answer: Feature scaling brings numerical features to comparable ranges. It is important for algorithms such as KNN and K-Means because they use distance calculations. Without scaling, a feature with a large numerical range may dominate other features.

14. What is normalization?

Answer: Normalization commonly rescales values to a fixed range, often 0 to 1. Min-max normalization uses (x–minimum)/(maximum–minimum). It is useful when features have very different ranges.

15. What is standardization? ★

Answer: Standardization transforms a feature by subtracting its mean and dividing by its standard deviation. The resulting feature has approximately zero mean and unit standard deviation. Unlike min-max normalization, values are not restricted to a fixed range.

16. Differentiate normalization and standardization.

Answer: Normalization usually maps values to a fixed range such as 0 to 1. Standardization centers the feature around zero and scales it using standard deviation. The choice depends on the algorithm and the data distribution.

17. What is an outlier?

Answer: An outlier is a data point that is unusually far from most other observations. It may be caused by an error, or it may represent a genuine rare event. Outliers should be investigated before deciding whether to remove or retain them.

18. What is feature selection?

Answer: Feature selection chooses the most useful original input features and removes irrelevant or redundant ones. It can reduce training time, simplify the model and sometimes improve generalization. It is different from creating new transformed features.

19. What is dimensionality reduction?

Answer: Dimensionality reduction decreases the number of input variables while trying to preserve important information. It can reduce computation, remove redundancy and help visualize high-dimensional data. PCA is a common dimensionality-reduction technique.

20. What is PCA? ★

Answer: Principal Component Analysis transforms the original features into new uncorrelated variables called principal components. The first component captures the largest possible variance, and later components capture remaining variance. PCA is commonly used for dimensionality reduction and visualization.

3. Regression and Gradient Descent

21. What is regression? ★

Answer: Regression is a supervised learning task used to predict a continuous numerical value. The model learns the relationship between input features and a numeric target. Examples include predicting Uber fares, temperature or house prices.

22. What is Linear Regression? ★

Answer: Linear Regression models the relationship between input features and a continuous target using a linear equation. It learns coefficients that minimize prediction error. It is simple, interpretable and commonly used as a baseline regression model.

23. Write the equation of simple Linear Regression.

Answer: The equation is y = mx + c. Here, x is the input, y is the predicted output, m is the slope and c is the intercept. The model learns m and c from the training data.

24. What is the cost function in regression?

Answer: A cost function measures how far the model's predictions are from the actual values. Mean Squared Error is a common cost function for Linear Regression. Training attempts to find parameter values that minimize this cost.

25. What is Mean Squared Error (MSE)? ★

Answer: MSE is the average of the squared differences between actual and predicted values. Squaring makes larger errors contribute more strongly to the total. A lower MSE generally indicates better predictions on the evaluated dataset.

26. What is Mean Absolute Error (MAE)?

Answer: MAE is the average of the absolute differences between actual and predicted values. It is expressed in the same units as the target, which makes it easy to interpret. It is less sensitive to large errors than MSE.

27. What is RMSE?

Answer: Root Mean Squared Error is the square root of Mean Squared Error. It is expressed in the same units as the target variable. Because errors are squared before averaging, RMSE penalizes large errors more strongly than MAE.

28. What is R-squared? ★

Answer: R-squared measures how much of the variation in the target is explained by the model compared with predicting the target mean. A value closer to 1 often indicates a better fit. It can be negative on test data when the model performs worse than the mean baseline.

29. What is Gradient Descent? ★

Answer: Gradient Descent is an optimization algorithm used to minimize a cost or loss function. It repeatedly updates model parameters in the direction opposite to the gradient. The process continues until the cost stops improving significantly or a stopping condition is reached.

30. Write the Gradient Descent update rule.

Answer: The update rule is $\theta = \theta - \alpha \times$ gradient of the cost with respect to θ. Here, θ represents a model parameter and α is the learning rate. The negative gradient direction is used because it reduces the cost locally.

4. Classification and K-Nearest Neighbours

31. What is classification? ★

Answer: Classification is a supervised learning task in which a model predicts a class or category. Examples include spam versus non-spam email and diabetic versus non-diabetic prediction. The output is discrete rather than a continuous number.

32. What is K-Nearest Neighbours (KNN)? ★

Answer: KNN is a supervised learning algorithm that predicts the class of a new sample using the labels of its K closest training samples. It commonly uses majority voting for classification. It is called a lazy learner because it stores training data rather than building a complex model during training.

33. Explain the working steps of KNN. ★

Answer: First choose the value of K and calculate the distance from the test sample to training samples. Select the K nearest samples and check their labels. The most frequent class among those neighbours becomes the predicted class.

34. What is the role of K in KNN?

Answer: K represents the number of neighbours considered for prediction. A very small K can be sensitive to noise, while a very large K can make the model too general. The value is usually selected using validation or cross-validation.

35. Which distance measures are used in KNN?

Answer: Euclidean distance is commonly used for numerical features and calculates straight-line distance. Manhattan distance calculates the sum of absolute differences. The selected distance measure can affect the classification result.

36. Why is feature scaling important in KNN? ★

Answer: KNN uses distances to identify nearby samples. If one feature has a much larger range than others, it can dominate the distance calculation. Scaling makes the features more comparable and helps the algorithm consider them fairly.

37. What are the time and space complexities of KNN?

Answer: For n training samples and d features, a direct prediction calculates distances in O(nd) time. Sorting all distances can add O(n log n), although selection methods can avoid a full sort. The stored training data requires O(nd) space.

38. What is a confusion matrix? ★

Answer: A confusion matrix compares actual class labels with predicted labels. For binary classification it contains true positives, true negatives, false positives and false negatives. It helps identify which types of prediction errors the model makes.

39. Define accuracy, precision and recall. ★

Answer: Accuracy is the fraction of all predictions that are correct. Precision is TP/(TP+FP), showing how many predicted positives are actually positive. Recall is TP/(TP+FN), showing how many actual positives the model finds.

40. What is F1-score?

Answer: F1-score is the harmonic mean of precision and recall. Its formula is 2 × (precision × recall)/(precision + recall). It is useful when both false positives and false negatives matter, especially with imbalanced classes.

5. Clustering and K-Means

41. What is clustering? ★

Answer: Clustering is an unsupervised learning technique that groups similar data points together. It does not require known target labels. It is used in customer segmentation, document grouping and pattern discovery.

42. What is K-Means clustering? ★

Answer: K-Means divides data into K clusters, with each cluster represented by a centroid. Each data point is assigned to the nearest centroid, and centroids are recalculated using the mean of assigned points. These steps repeat until the clusters stop changing significantly.

43. Explain the steps of the K-Means algorithm. ★

Answer: Choose the number of clusters K and initialize K centroids. Assign every data point to its nearest centroid, then calculate new centroids as the mean of assigned points. Repeat assignment and update until convergence or a maximum number of iterations is reached.

44. What is a centroid?

Answer: A centroid is the mean position of all points assigned to a cluster. It represents the centre of that cluster in feature space. K-Means updates centroids after each assignment step.

45. What is inertia in K-Means?

Answer: Inertia is the sum of squared distances between each data point and its assigned centroid. Lower inertia means points are closer to their cluster centres. However, inertia generally decreases when K increases, so it should not be used alone to choose K.

46. What is the Elbow Method? ★

Answer: The Elbow Method runs K-Means for different values of K and plots the inertia for each value. Inertia decreases as more clusters are added. The point where the decrease starts slowing down, forming an elbow, can be a reasonable choice for K.

47. What are the time and space complexities of K-Means?

Answer: A common time estimate is O(nKdi), where n is the number of samples, K is the number of clusters, d is the number of features and i is the number of iterations. Data storage is O(nd), while centroids require O(Kd) and assignments O(n).

48. Why should features be scaled before K-Means? ★

Answer: K-Means uses distance calculations to assign points to centroids. Features with larger numerical ranges can dominate these distances. Scaling prevents one feature from controlling the clustering only because of its units.

49. What are the limitations of K-Means?

Answer: The number of clusters K must be selected in advance, and the result can depend on initial centroids. K-Means is sensitive to outliers and feature scaling. It works best when clusters are roughly compact and similarly sized.

50. Differentiate K-Means and KNN.

Answer: K-Means is an unsupervised clustering algorithm that groups unlabelled data. KNN is a supervised algorithm that predicts labels using nearby labelled examples. K-Means learns centroids, while KNN mainly stores the training data.

6. Model Evaluation and Generalization

51. What is overfitting? ★

Answer: Overfitting happens when a model learns the training data too closely, including noise and accidental patterns. It may perform very well on training data but poorly on unseen data. More data, simpler models and regularization can help reduce overfitting.

52. What is underfitting? ★

Answer: Underfitting occurs when a model is too simple or has not learned enough to capture important patterns. It performs poorly on both training and testing data. A more suitable model or better features may improve performance.

53. Explain the bias-variance trade-off.

Answer: Bias is error caused by overly simple assumptions, while variance is sensitivity to changes in training data. High bias can cause underfitting and high variance can cause overfitting. A good model balances both to generalize well.

54. What is generalization in Machine Learning?

Answer: Generalization is the ability of a trained model to make accurate predictions on new data that it has not seen before. A model that only memorizes its training examples has poor generalization. Testing and cross-validation help estimate generalization.

55. Why can accuracy be misleading for an imbalanced dataset?

Answer: If one class is much more common, a model can predict only that class and still achieve high accuracy. However, it may fail to identify the minority class. Precision, recall, F1-score and the confusion matrix give more information.

56. What is the difference between micro and macro averaging?

Answer: Macro averaging calculates a metric separately for each class and then takes the simple average, giving every class equal weight. Micro averaging combines the total true positives, false positives and false negatives across classes before calculating the metric. They can give different results when class sizes are unequal.

57. What is regularization?

Answer: Regularization adds a penalty to the model's objective to discourage overly complex parameter values. It can reduce overfitting and improve generalization. L1 and L2 regularization are common examples.

58. Differentiate Ridge and Lasso Regression.

Answer: Ridge Regression uses an L2 penalty, which shrinks coefficients but usually does not make them exactly zero. Lasso uses an L1 penalty and can make some coefficients zero, performing feature selection. Both help control overfitting.

59. What is a validation dataset?

Answer: A validation dataset is used during model development to compare models and tune hyperparameters. It is separate from the training data used to fit model parameters. The test dataset should be kept for final evaluation.

60. What is a hyperparameter? Give examples.

Answer: A hyperparameter is a setting chosen before or during model selection rather than learned directly as a model coefficient. Examples include K in KNN, number of clusters in K-Means and learning rate in Gradient Descent. Hyperparameters can be tuned using validation data.

7. Practical Viva – Uber Fare Prediction

61. Is Uber fare prediction a classification or regression problem? ★

Answer: It is a regression problem because the output is a continuous numerical fare amount. The model learns the relationship between trip-related features and the fare. The predicted fare can be compared with the actual fare using regression metrics.

62. What features can be used for Uber fare prediction?

Answer: Possible features include pickup and drop-off coordinates, trip distance, passenger count and pickup date or time. The exact features depend on the dataset supplied for the practical. The fare amount is generally the target variable.

63. Why should invalid latitude and longitude values be removed?

Answer: Latitude and longitude must fall within valid geographic ranges. Incorrect coordinates can create unrealistic distances and affect model training. Data cleaning helps prevent such errors from influencing predictions.

64. How can distance between pickup and drop-off locations be calculated?

Answer: The Haversine formula estimates the great-circle distance between two latitude-longitude points on Earth. It uses the coordinates and Earth's radius. This is more appropriate for geographic distance than applying ordinary Euclidean distance directly to raw coordinates.

65. Why can pickup time be useful for fare prediction?

Answer: Demand and fares may vary by hour, weekday, season or peak travel period. Time values can be converted into features such as hour, day or month. These features may help the model learn time-based patterns if the dataset contains them.

66. Which metrics can be used to evaluate Uber fare prediction? ★

Answer: MAE, MSE, RMSE and R-squared are commonly used regression metrics. MAE and RMSE show prediction error in fare units, while R-squared measures explained variation. It is useful to compare more than one metric.

67. Why do we use train_test_split in this practical?

Answer: It divides the dataset into training and testing parts. The model learns from the training part and is evaluated on the testing part. This gives an estimate of how well the model predicts fares for unseen trips.

68. What problems can occur if the fare dataset contains outliers?

Answer: Unusually high fares, incorrect coordinates or unrealistic trip distances can influence the model strongly. Such values may increase prediction error, especially for models sensitive to large errors. They should be checked before deciding whether to remove or retain them.

8. Practical Viva – Email Spam Classification

69. Is email spam detection a classification problem? ★

Answer: Yes, it is usually a binary classification problem. The model predicts whether an email is spam or not spam. The output is a category rather than a numerical value.

70. Why is text preprocessing required for spam detection?

Answer: Raw email text may contain inconsistent capitalization, punctuation and unnecessary tokens. Preprocessing can make the text more consistent before converting it into numerical features. The exact steps depend on the notebook's text-processing pipeline.

71. What is tokenization?

Answer: Tokenization splits a text document into smaller units called tokens, often words or subwords. These tokens can then be counted or transformed into numerical features. It is a basic step in many text-processing workflows.

72. Why must text be converted into numbers before classification?

Answer: Most traditional Machine Learning algorithms work with numerical feature vectors rather than raw text. Text vectorization converts words or word patterns into numbers. The classifier then learns patterns from these numerical representations.

73. What is the Bag of Words model? ★

Answer: Bag of Words represents a document using the frequency or presence of words. It ignores the order of words and focuses on which words occur and how often. It is a simple way to convert text into numerical features.

74. What is TF-IDF?

Answer: TF-IDF stands for Term Frequency–Inverse Document Frequency. It gives higher importance to words that are frequent in one document but less common across all documents. It can reduce the influence of very common words.

75. What is a false positive in spam detection?

Answer: A false positive occurs when a genuine email is incorrectly classified as spam. This is important because the user may miss an important message. Precision is useful for measuring how reliable spam predictions are.

76. Which evaluation metrics are useful for spam detection? ★

Answer: Accuracy, precision, recall, F1-score and the confusion matrix are useful. Precision matters when we want to avoid marking genuine emails as spam. Recall matters when we want to catch as many spam emails as possible.

9. Practical Viva – Gradient Descent

77. What is the aim of the Gradient Descent practical? ★

Answer: The aim is to minimize a cost function by repeatedly updating model parameters. The gradient indicates how the cost changes when parameters change. The algorithm moves in the opposite direction of the gradient to reduce the cost.

78. What are the parameters in simple Linear Regression?

Answer: The main parameters are slope or weight m and intercept or bias c. The model learns these values from the training data. They define the line used to predict the output from an input.

79. What does the gradient tell us?

Answer: The gradient tells us the direction and rate of the steepest increase in the cost function. To reduce the cost, Gradient Descent moves in the opposite direction. The size of the movement is controlled by the learning rate.

80. What happens if the learning rate is too high? ★

Answer: A very high learning rate can make parameter updates too large. The algorithm may jump over the minimum, oscillate or even diverge. The cost may fail to decrease.

81. What happens if the learning rate is too low?

Answer: A very low learning rate makes each parameter update very small. The algorithm may take a large number of iterations to reach a good solution. Training becomes slow even if it eventually converges.

82. What is convergence in Gradient Descent?

Answer: Convergence means the parameter values or cost have become stable enough that further updates make very little difference. A program may stop when the cost change is small or when a maximum number of iterations is reached. The stopping rule depends on the implementation.

83. Why do we plot the cost function during Gradient Descent?

Answer: A cost-versus-iteration plot shows whether the algorithm is reducing error over time. A decreasing curve usually indicates progress toward a minimum. If the cost increases or oscillates heavily, the learning rate or implementation may need adjustment.

84. Differentiate Batch and Stochastic Gradient Descent.

Answer: Batch Gradient Descent calculates the gradient using the full training dataset for each update. Stochastic Gradient Descent uses one training example per update. Batch updates are generally smoother, while stochastic updates are more frequent and noisy.

10. Practical Viva – KNN on Diabetes Dataset

85. What is the objective of the KNN diabetes practical? ★

Answer: The objective is to classify a person into a target category, typically diabetic or non-diabetic, using the available health-related input features. KNN compares the person with similar examples in the training data. The exact target column depends on the dataset used in the notebook.

86. Why is feature scaling important in the diabetes KNN practical? ★

Answer: Features such as glucose, age and BMI may have different numerical ranges. KNN calculates distances, so a large-range feature can dominate the result if data is not scaled. Scaling makes the distance calculation more balanced.

87. Why should the dataset be split before training and testing?

Answer: The model should be evaluated on examples that were not used to train it. The training set is used to store or learn patterns, while the test set measures performance on unseen samples. This helps estimate real-world generalization.

88. Why should the scaler be fitted only on training data?

Answer: The scaler learns values such as the mean and standard deviation from the data. If it is fitted on the full dataset, information from the test set influences preprocessing. Fit it on training data and then transform both training and test sets.

89. How does KNN predict the diabetes class?

Answer: It calculates the distance from the test person's feature values to the training samples. It selects the K nearest samples and checks their class labels. The majority class among those neighbours becomes the prediction.

90. How can we select a suitable value of K?

Answer: Try different K values and evaluate them on validation data or using cross-validation. Compare metrics such as recall, precision and F1-score, not only accuracy. Choose the value that performs well on unseen data.

91. Why can accuracy be misleading in diabetes prediction?

Answer: If one class is more common, a model may achieve high accuracy by predicting that class most of the time. It may still fail to identify many positive cases. Recall, precision, F1-score and the confusion matrix provide a fuller picture.

92. What happens when K=1 in KNN?

Answer: The predicted class is taken from the single nearest training sample. This can make the model very sensitive to noise or an incorrectly labelled sample. It may overfit the training data.

11. Practical Viva – K-Means Clustering

93. Why is K-Means called an unsupervised algorithm? ★

Answer: K-Means does not require target labels during training. It groups data points based on their similarity in the selected features. The cluster numbers are created by the algorithm and do not automatically represent meaningful real-world categories.

94. What does fit() do in a K-Means implementation?

Answer: The fit() method runs the clustering algorithm on the supplied feature data. It learns the cluster centroids and assigns training samples to clusters. The exact behaviour depends on the library implementation and its parameters.

95. What does predict() return in K-Means?

Answer: For a fitted K-Means model, predict() assigns each supplied sample to the nearest learned centroid. It returns a cluster index or label for each sample. These labels identify clusters, not pre-existing target classes.

96. What is random_state used for?

Answer: random_state controls random operations such as centroid initialization or data splitting. Setting it helps make results reproducible when the same code is run again. It does not guarantee that a different dataset or library version gives identical results.

97. What is n_init in K-Means?

Answer: n_init specifies how many times K-Means is run with different initial centroid choices. The implementation selects the run with the best clustering objective, commonly the lowest inertia. Multiple initializations reduce the chance of a poor result from one starting point.

98. What is convergence in K-Means?

Answer: K-Means converges when cluster assignments or centroid positions stop changing significantly, or when a stopping limit is reached. At that point, further iterations are unlikely to improve the result much. The algorithm may also stop after a maximum number of iterations.

99. How do you interpret cluster labels?

Answer: Cluster labels such as 0, 1 and 2 are identifiers assigned by the algorithm. They do not mean that cluster 2 is better or larger than cluster 1. To understand them, examine the feature values or characteristics of the points in each cluster.

100. Give real-life applications of K-Means.

Answer: K-Means can be used for customer segmentation, grouping similar documents, image colour compression and identifying patterns in sensor data. For example, a business can group customers by purchasing behaviour. The clusters can then support targeted decisions.